

Srikrishna Bangalore Raghu

Doctoral Student in Computer Science

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Summary

My research develops learning-based motion generation models and sampling-based kinodynamic planners, deployed on real manipulator hardware. I am seeking a Summer 2027 robotics internship in motion planning and robot learning.

Education

University of Colorado Boulder (Boulder, CO, USA)

2023–Present

Ph.D. in Computer Science (GPA 3.889/4) — Expected 2028

PES University (Bengaluru, India)

2017–2021

B.Tech in Electrical and Electronics Engineering (CGPA 8.78/10)

Research Experience

PhD Student, HIRO Group, University of Colorado Boulder, Advisor: Alessandro Roncone

Aug 2023–Present

- Designed and implemented Bidirectional Bundle of Edges (BBoE), a bidirectional kinodynamic motion planner in Python, producing solutions 1.5x faster than state-of-the-art planners in cluttered environments.
- Developed a hesitancy-conditioned diffusion model (PyTorch) that generates real-world Franka Emika Panda trajectories at a commanded level of hesitancy.
- Built the collection and labeling pipeline for Dance2Hesitate, a 220-trajectory multimodal dataset of hesitant motion (synchronized 30 Hz robot-state streams from Franka kinesthetic teaching), released open source on [Hugging Face](#). [Link](#)
- Trained Transformer and LSTM decoders that classify hesitancy intensity from motion (leave-one-dancer-out $\kappa = 0.76$ on unseen performers) and ran an in-person perception study ($N = 33$).
- Generated emotionally and functionally expressive manipulator trajectories using Laban Movement Analysis (LMA). Human trials supported 2 of 3 hypotheses. [Link](#)

Research Assistant, Artificial Intelligence and Robotics Lab, Indian Institute of Science

Jul 2021–Jun 2022

- Implemented a heterogeneous multi-task multi-robot task allocation algorithm for TurtleBot3 systems, deployed in simulation and on real robots via ROS 2 with Gmapping SLAM.

Publications

- S. Bangalore Raghu**, A. Soukhovei, D. Vankineni, A. Roncone. *What makes Robot Motion Hesitant? Producing and Perceiving Hesitancy in Robotic Manipulators*. **ACM/IEEE International Conference on Human-Robot Interaction (HRI)**, 2027. (Submitted).
- S. Bangalore Raghu**, A. Soukhovei, D. Vankineni, A. Bacula, A. Roncone. *Dance2Hesitate: A Multi-Modal Dataset of Dancer-Taught Hesitancy for Understandable Robot Motion*. **Designing Transparent and Understandable Robots (D-TUR) Workshop at the International Conference on Human-Robot Interaction (HRI)**, 2026. [Link](#)
- S. Bangalore Raghu**, C. Lohrmann, A. Bakshi, J. Kim, J. C. Herrera, B. Hayes, A. Roncone. *Employing Laban Shape for Generating Emotionally and Functionally Expressive Trajectories in Robotic Manipulators*. **IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)**, 2025. [Link](#)
- S. Bangalore Raghu**, A. Roncone. *BBoE: Leveraging a Bundle of Edges for Kinodynamic Bidirectional Motion Planning*. **Robotics and Automation Letters (RA-L)**, 2026. (Submitted). [Link](#)
- S. B. Velhal, **S. Bangalore Raghu**, M. Bharatheesha, S. Sundaram. *A Dynamic Heterogeneous Team-based Non-iterative Approach for Online Pick-up and Just-In-Time Delivery Problems*. **arXiv**, 2024. (Preprint). [arXiv](#)

Awards & Teaching

Departmental Summer Research Fellowship, University of Colorado Boulder

2026

Published Work Award, Research Expo, University of Colorado Boulder

2026

Graduate Teaching Assistant (CSCI 1300, 2270, 3302; ROBO 5000), University of Colorado Boulder

2023–Present

Skills

Programming: C++, Python, C, MATLAB, Java, R, Julia

Hardware: Franka Emika Panda, TurtleBot3

Robotics/ML: Motion Planning, Diffusion Models, Transformers, Deep Learning, Reinforcement Learning, HRI

Frameworks/Tools: PyTorch, ROS/ROS 2, Linux, MoveIt, libfranka, Gazebo, PyBullet, Webots, OpenCV, Docker, Git, AWS